A Synopsis

Of

Project

On

EEG-ECG Analysis of Physiological Differences Associated with Yoga Experience

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING (with specialization in Artificial Intelligence & Data Science)

Under the Guidance of

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SEPTEMBER 2026

CERTIFICATE

This is to certify that the project synopsis entitled **"EEG-ECG Analysis of Physiological Differences Associated with Yoga Experience"** is being submitted by **Dev Goyal (GE-232023857) and Nandini Aggarwal (GE-232023893)** for the partial fulfillment of the requirement for the award of the degree of **B. Tech-CSE (with specialization in Artificial Intelligence & Data Science)** in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the synopsis has not been submitted in any other university or institution for the award of any degree.

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DECLARATION

We hereby declare that the project synopsis entitled "**EEG-ECG Analysis of Physiological Differences Associated with Yoga Experience"** is being submitted for the partial fulfillment of the requirement for the award of the degree of **B. Tech-CSE (with specialization in Artificial Intelligence & Data Science)** in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

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Abstract

Changes to both neural and autonomic physiological activity are tied closely with mental health and psychological well-being. Yoga and meditation practices have been increasingly studied as possible regulators of stress, attention and cognitive performance, nevertheless, the vast majority of existing studies have focused on EEG vs cardiac measures (ECG and HRV) in isolation. This limits our understanding of the complementary information provided by the brain and heart. The objective of this project is to characterize mental-health-related physiological differences between experienced yoga practitioners and non-practitioners using a proposed multimodal EEG-ECG framework.

The proposed methodology analyzes synchronized EEG and ECG recordings across different cognitive states: resting state, internal concentration, external concentration, and mind-wandering. After that a systematically comparison of three representation strategies: EEG-only, ECG/HRV-only and combined EEG-ECG is conducted. Besides traditional multimodal feature fusion, the study will explore brain-heart interactions, which examines whether the association between neural and cardiac activity is conditioned by yoga experience and cognitive state. Furthermore, the study proposes to extract and analyze features like spectral characteristics of EEG, HRV measures, and appropriate EEG-ECG coupling measures.

The study also compares several machine learning methods (Logistic Regression, Support Vector Machine, Random Forest and XGBoost) to identify which strategy optimally distinguishes these physiologic differences. In the study, subject-independent validation is used to minimize leakage and to provide a more honest evaluation of generalization. In addition, unsupervised K-Means clustering will be applied to find physiological profiles without using practitioner labels to create clusters. Additionally, focused yoga-intervention studies may provide supporting evidence for pre-post physiological changes. The potential effects of brain-heart coupling on the understandings of yoga-induced differences in mental health and cognitive functioning are examined in systematic comparison with unimodal vs. multimodal physiological representations within proposed work.

1. Introduction

Psychological well-being and mental health are mediated by neural and autonomic physiological activity. This also implies that EEG, ECG and HRV may provide complementary data in the study of stress, cognition and mental states. The utility of a multimodal approach in physiological assessment was highlighted by a study using EEG, ECG and cognitive tests to show measurable neural and cardiac changes after Sudarshan Kriya Yoga [1].

In addition, data from EEG measurement and ECG measurement have also confirmed their complementary nature by showing that features obtained from mixing both measurements were better in stress assessment than each separate feature set [2]. Likewise, HRV measures have been employed to study the alterations of cardiac autonomic function which is related to structured practice of Hatha Yoga [5].
This combined evidence suggests both neural and cardiac activity may contain information about yoga-related changes in physiology.

However, the rising application of machine learning-based methodologies for physiological-signal analysis has still encountered various challenges, including small datasets, subject-to-subject variability, data leakage, inconsistent preprocessing and insufficient cross-subject validation [8]. Reviews of open physiological datasets have also highlighted challenges with synchronization and heterogeneous protocols [9]. These challenges emphasize the importance of controlled multimodal analysis and subject-independent evaluation.

Recent work has shown that EEG-ECG fusion offers significant information in this domain, since attention-based multimodal models outperform the same model with only EEG or ECG when classifying stress [13]. But almost all of the yoga and meditation research has looked at these practices separately. EEG studies have revealed reliable neural correlates of meditation expertise [14], but do not concurrently record cardiac responses.

The EEGMeditation dataset, which has synchronized EEG and ECG recordings of yoga practitioners and non-practitioners while performing tasks of concentration and mind-wandering [15]. This allows for direct comparison of neural and cardiac activity across yoga experience and cognitive states. But even if it provides the most informative representation, data alone cannot dictate whether EEG, ECG/HRV or a combination of both yields more information [15].

Multimodal fusion also does not represent the interaction between these two physiological systems. Limited evidence from concurrent EEG-ECG studies suggests considerable variability in methods [17]. On the other hand, machine-learning investigations have shown that subject-independent models can appropriately characterize EEG patterns related to Himalayan Yoga meditation [18].

Thus, based on these results, the study offers a multimodal EEG-ECG framework for the characterization of physiological differences that relate to mental health and are associated with experience in yoga. This study will also compare EEG-only, ECG/HRV-only, and combined representations of the brain and heart signal across cognitive states (brain-heart coupling analysis). Subject-independent validation will be conducted for multiple machine-learning models, while K-Means will also be investigated to identify physiological profiles with no practitioner labels [8] [9] [18]. This study aimed to examine whether combinations of brain and heart activity and their interactions characterize familiar forms of autonomic regulation that are distinctively associated with yoga, which would provide a fuller portrayal of the functional correlates underlying the mental health and cognitive benefits previously reported [1] [13] [15] [17].

2. Literature Review

In recent years, increased research on yoga and meditation has begun to investigate their effects on neural, autonomic, and psychological function. An experiment conducting an early investigation of Sudarshan Kriya Yoga in Adult participants using EEG, ECG, and cognitive testing provided descriptive evidence for changes in neural and cardiac measures following the intervention, suggesting that yoga-related

autonomic changes may occur through interactions between the brain and heart [1]. Following such multimodal vision, a wearable EEG-ECG system for assessing stress was created and combination of EEG and HRV features improved classification accuracy versus activation of either signal only [2]. These results suggest that neural and cardiac signals can provide complementary information in terms of the physiological response associated with a psychological state being examined.

Research on broader mind-body practices have suggested possible effects on neural and autonomic dynamics of techniques such as yoga, meditation, Tai Chi, Qigong, and breathwork, as well as the increasing number of physiological monitoring technologies such EEG and HRV [3]. In addition, multimodal physiological research has developed a deep-learning framework for modelling temporal changes in psychological states integrating EEG, HRV, EDA and other signals [4]. These types of approaches highlight a growing transition away from purely physiological measures, towards multimodal representations of psychological and emotional states.

Yoga research on heart activity using HRV measures in a structured Hatha Yoga intervention, RMSSD, SDNN, LF, HF and LF/HF were investigated to assess changes in cardiac function [5]. This indicates the relevance of ECG-related parameters in characterizing physiological change during yoga. Nonetheless, cardiac metrics cannot directly reflect associated shifts in neural activity, suggesting that combining cardiac and brain measurements may provide more information.

Studies based on physiological indicators have explored neural and autonomic responses linked to stress, anxiety and emotional regulation. A systematic review examined various physiological measures, including EEG, ECG/HRV, fMRI, fNIRS, EDA, EMG, and biochemical markers, in relation to psychological-state assessment [6]. In a similar way, EEG and ECG derived biomarkers have been found in clinical studies to predict and suggest treatment plans and strategy for depressed patients [7]. While these studies provide evidence that physiological measures can reflect pathology, they focus on different populations and research goals than studies examining physiological differences associated with yoga experience.

Machine learning provides another way to analyze physiological signals. A systematic review examined the use of EEG, ECG/HRV, EDA, and other physiological signals for assessing stress and anxiety using machine learning [8]. It also highlighted several challenges, including small datasets, differences between participants, variations within the same participant, data leakage between training and test sets, inconsistent preprocessing, and limited cross-subject validation. These issues are particularly important for physiological data because many signal segments may come from only a small number of participants.

Another study that conducted a systematic review on open physiological datasets listed additional challenges due to heterogeneous experimental protocols, sensor variability (placement, orientation), synchronization, labelling and class imbalance [9]. These limitations show the need for careful experimental comparisons and subject-independent evaluation of machine-learning models. Multimodal representations have also been studied aside from physiological signals alone; for example, EEG-derived features and behavioral measures with personality characteristics were used with deep-learning architectures to recognize transitions of dynamic emotion states [10]. This kind of work highlights the trend towards learned representations for difficult neural network-led psychological-state modelling but does not speak directly to yoga-associated brain-heart physiology.

Advanced meditation research has also shown multimodal physiological recording. In one of the case studies, simultaneous EEG, ECG, EMG and respiration were recorded during deep meditative state in a real-life setting showing that multiple physiological systems can be observed simultaneously with advanced yogic practice [11]. Recent research has explored the relationship between cardiac and brain activity in using heart rate variability (HRV) and functional neuroimaging methods, such as functional magnetic resonance imaging (fMRI), showing that temporal heart-brain coupling can show psychological and emotion-regulation processes [12]. However, it uses fMRI rather than EEG and does not examine differences between yoga practitioners and non-practitioners.

Previous work using machine-learning approaches have provided more convincing evidence for the advantages of multimodal physiological fusion. Combined modalities: An attention-based EEG-ECG framework improved stress-classification performance compared to using estimation of EEG or ECG independently [13] Such results indicate that multimodal fusion can capture complementary information from both neural and cardiac signals. However, the study was performed on general stress datasets instead of a yoga practitioner versus non-practitioner population.

In respect to research on meditation, EEG has been applied really successfully to identify repeated neural states related to levels of skilled meditative practice [14]. Importantly, while these findings show that EEG can reflect meaningful differences associated with extensive meditation experience, the analysis of EEG alone does not permit conclusions about whether similar patterns of autonomic change were present or how neural and cardiac activity may become differentially coordinated with practice.

Specifically, there is a synchronized multimodal dataset that provides EEG and ECG recordings from yoga practitioners and non-practitioners across concentration and mind-wandering conditions [15]. This dataset represents a critical step towards fully investigating neural and cardiac activity in the same participants during multiple cognitive states. Yet, the actual dataset does not confirm which physiological system (EEG, ECG/HRV or their combination) would represent more confident data nor explores directly the relation between both systems.

EEG and ECG have also been studied together. One study proposed a method to measure the complexity and interaction between signals from both modalities simultaneously [16]. These methods show that there can be not only information in the individual physiological signals but also in their joint dynamics. This provides support for the wider motivation for investigating relationships between brain and cardiac activity as opposed to using exclusively independent modality specific features.

However, direct brain-heart coupling remains under-explored. A review found that only about 5% of EEG-ECG studies directly assessed brain-heart coupling [17]. This distinction is significant since traditional feature fusion (e.g. simply concatenating EEG and ECG features) does not necessarily reflect any temporal or functional relationship between the two systems.

Machine learning studies of meditation-related EEG signals have shown that meditative groups can be classified using deep-learning models while keeping participants separate between training and testing [18]. This emphasizes the opportunity to leverage machine learning for physiological performance analysis in yoga and the need for robust subject-independent assessment.

Collectively, the literature suggests that physiological responses to yoga and meditation cannot be completely captured multi-dimensionality by a one-dimensional perspective. EEG studies reflect alterations in brain activity and meditation-related states, while ECG/HRV studies provide additional information of autonomic regulation [1] [5] [14] [18]. Concurrently, multimodal studies demonstrated the potential for enhancing characterization of stress and psychological states by combining neural and cardiac information [2] [13].

However, previous studies either used a single physiological signal or combined signals in settings unrelated to yoga experience [4] [6] [7] [10]. Additionally, the methodology literature suggests that differences in experimental protocols, synchronization, sample size and validation strategies may greatly impact the reliability of physiological machine-learning findings [8] [9].

3. Problem Statement and Objectives

3.1 Problem Statement

Previous research show that yoga and meditation can affect brain activity and cardiac autonomic regulation. EEG and ECG/HRV are often studied separately, while recent studies have also compared experienced yoga practitioners with non-practitioners. However, multimodal studies usually

combine EEG and ECG features without examining whether the interaction between brain and cardiac activity provides additional information. Another limitation is the lack of synchronized EEG and ECG analysis across different cognitive states in yoga practitioners and non-practitioners. Therefore, this study aims to compare the information provided by EEG, ECG/HRV, their combination, and their interaction in identifying physiological differences associated with yoga experience. It will also examine whether these differences remain consistent across cognitive states and can be reliably detected using subject-independent machine-learning models.

3.2 Objectives

- To analyze EEG-based physiological characteristics of experienced yoga practitioners and non-practitioners across different cognitive states, including resting, internal concentration, external concentration, and mind-wandering conditions.
- To extract and analyze ECG/HRV-based characteristics to investigate differences in cardiac autonomic activity associated with yoga experience.
- To systematically compare EEG-only, ECG/HRV-only, and combined EEG–ECG representations to determine the relative contribution of each physiological modality.
- To investigate brain–heart interactions using suitable EEG–ECG coupling measures and determine whether the relationship between neural and cardiac activity differs with yoga experience and cognitive state.
- To evaluate multiple machine-learning approaches, including Logistic Regression, Support Vector Machine, Random Forest, and XGBoost, for identifying physiological patterns associated with yoga experience.
- To employ subject-independent validation to reduce participant-level data leakage and provide a more reliable assessment of model generalization.
- To explore unsupervised K-Means clustering for identifying distinct physiological profiles without using practitioner/non-practitioner labels during cluster formation.

- To examine the consistency of physiological patterns across cognitive states, thereby determining whether observed differences are relatively stable or dependent on the individual's cognitive condition.

4. Proposed Methodology

4.1 Research Framework

The proposed project aims to determine whether EEG and ECG signals together provide more information about physiological differences related to mental health between experienced yoga practitioners and non-practitioners. The methodology compares the information from EEG only, ECG only, and their combined representation. It also examines whether these differences change across different cognitive states and how brain and cardiac activity are related.

The primary analysis will be conducted using the EEGMeditation dataset, which contains synchronized EEG and ECG recordings from experienced yoga practitioners and subjects with no prior experience of yoga or self-regulation practices, will be used for the main analysis. The recordings include multiple experimental conditions: resting state, mind-wandering (i.e., distracted), internal concentration, and external concentration. This allows us to compare practitioners and non-practitioners and examine whether the differences vary across cognitive states.

The complete processing pipeline consists of signal preprocessing, feature extraction, construction of unimodal and multimodal representations, brain-heart interaction analysis, machine learning, and statistical evaluation. The methodology is designed to maintain subject independence throughout model evaluation so that the obtained results reflect generalizable physiological patterns rather than participant-specific characteristics.

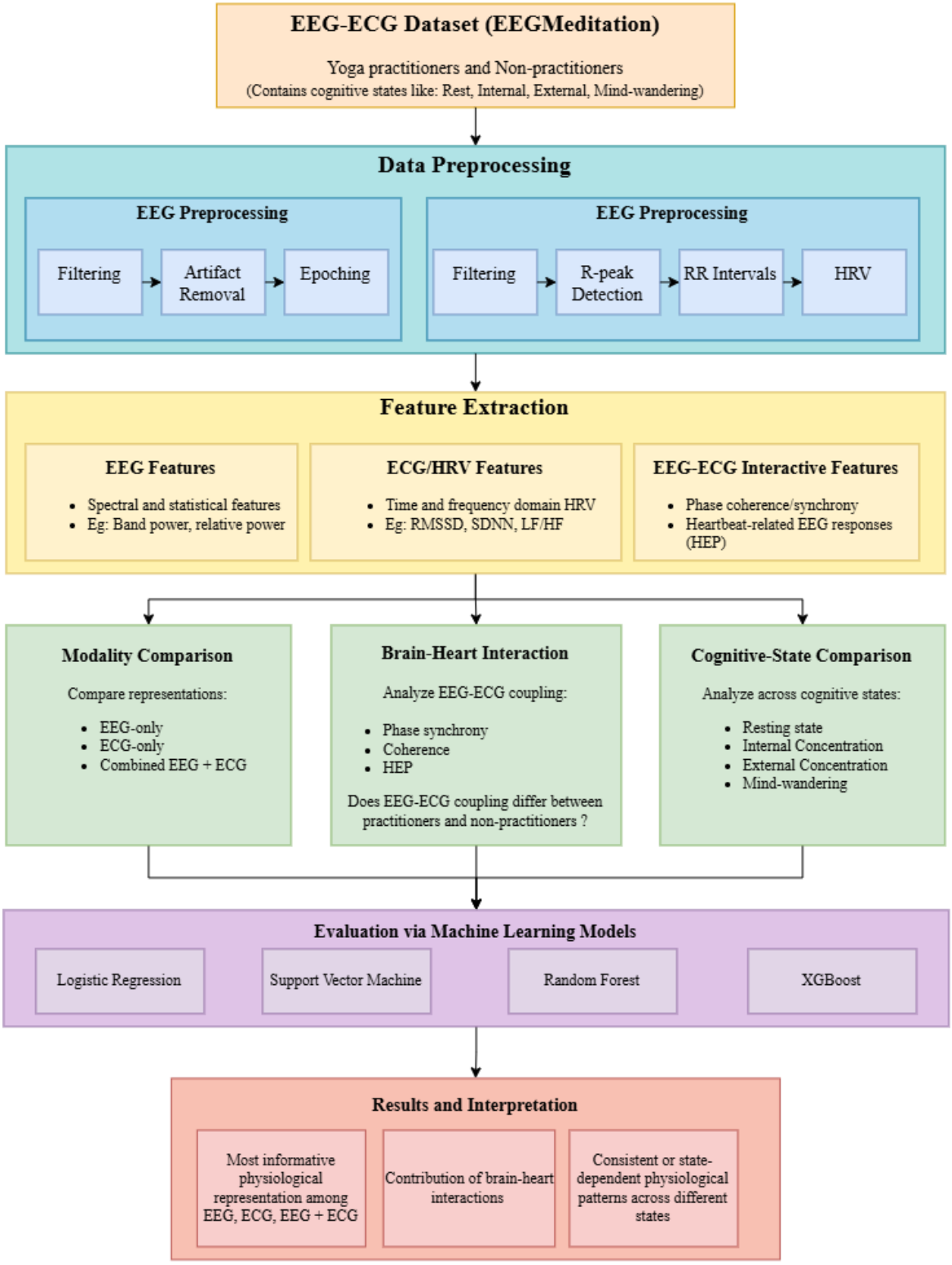


Figure 1. System architecture of the proposed EEG-ECG analysis framework.

4.2 Data Preprocessing

The study analyzes data quality of the EEG and ECG recordings in terms of signal quality, synchronization between datasets, missing data, and inconsistencies between recordings. Since both modalities are recorded together, the temporal alignment of the data will be validated to confirm that matching EEG and ECG segments correspond to physiological times.

Appropriate filtering of EEG signals will be performed to eliminate low-frequency drift, high-frequency noise and power line interference. Non-neural artifacts of eye-movements, muscle activity and other unknown sources will be detected and rejected applying appropriate artifact-rejecting techniques (e.g. Independent Component Analysis, signal quality evaluation). Segments labelled as bad channels or severely contaminated will be removed from further analysis steps.

Also, the ECG recordings will be filtered in a similar way as to remove baseline drifts and Kalman filter noise. Furthermore, the study will extract R-peaks from a clean ECG signal and get the inter-beat-intervals. They will be used to determine the basis of heart-rate and heart-rate-variability measures.

Once EEG and ECG signals are pre-processed in the original coordinate system, these will be split into suitable time windows. Each one of the time segments will have a matching part regarding the other modality (i.e. both modalities will consist of the same blocks) to allow for direct comparison and combination of features extracted from EEG and ECG. During machine learning evaluation, feature normalization will use statistics computed based on the training data only.

4.3 Feature Extraction and Multimodal Representation

EEG and ECG data will first be decomposed into derived features and then both modalities' representations will be sewn together. Frequency-domain features for EEG will include power of main frequency bands (delta, theta,

alpha, beta and gamma). Relative band power and selected band-power ratios can be computed as well. Statistical and nonlinear properties might be included such as variance, skewness, kurtosis or entropy-based measures.

ECG will be used to extract heart rate and standard HRV measures. These will consist of time domain indicators such as mean heart rate, SDNN and RMSSD, also frequency domain elements like LF, HF and LF/HF. Some nonlinear measures may be selected to capture changes in cardiac dynamics. Furthermore, three main feature representations will be constructed. The first will consist of only the EEG features, which represent information related to neural activity. The second will include ECG and HRV features only, which means cardiac autonomic information. The third will use feature-level fusion with both EEG and ECG features.

EEG-ECG interaction features will also be explored. These features will examine the relationship between brain and heart activity using measures such as correlations between EEG bands and HRV, temporal relationships, coherence, and heartbeat-related EEG responses when suitable signals are available. This analysis will examine whether brain and heart activity together provide more information than either modality alone.

4.4 Machine Learning and Evaluation

Supervised machine-learning algorithms will be used to evaluate the extracted representations and classify experienced yoga practitioners and non-practitioners. The models will include Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost.

All models will be tested separately using EEG-only, ECG-only, EEG+ECG, and EEG+ECG with interaction features. This will help compare the contribution of each modality and the brain–heart interaction features.

The evaluation will be subject-independent using group-based cross-validation. Data from the same participant will not be included in both training and testing sets. This is important because using multiple windows from the same participant in both sets can lead to overly high performance.

Model performance will be evaluated using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (if applicable). Performance will also be compared across cognitive states to examine whether the physiological differences are consistent across resting, concentration, and mind-wandering states.

4.5 Statistical and Unsupervised Analysis

The EEG, ECG, and brain-heart interaction features will be compared between practitioner and non-practitioner groups using statistical analysis and machine-learning methods. Effect sizes and appropriate statistical tests will also be used to understand the importance of the observed differences.

Another analysis will use K-Means clustering. EEG, ECG, and selected interaction features will be used to identify natural physiological profiles without providing practitioner or non-practitioner labels to the clustering algorithm. The resulting clusters will then be compared based on yoga experience and cognitive state. Cluster quality will be evaluated using measures such as the silhouette score.

Selected pre-and post-practice physiological measurements from yoga-intervention studies may also be used to support the interpretation of the observed physiological patterns. These analyses will be exploratory and will not be used to make causal or clinical claims.

Overall, the methodology will examine physiological differences between yoga practitioners and non-practitioners using EEG, ECG/HRV, multimodal

features, brain-heart interaction analysis, supervised machine learning, and unsupervised clustering.

5. Hardware and Software Requirements

The proposed work primarily involves processing synchronized EEG and ECG recordings, extracting physiological features, performing brain–heart coupling analysis, and training machine-learning models. The hardware requirement therefore consists of a computer with a multi-core processor, adequate RAM, and sufficient storage for handling multichannel physiological recordings. A GPU may be used if computationally intensive deep-learning experiments are included. The software implementation will primarily use Python with libraries such as MNE-Python for EEG processing, NumPy and SciPy for numerical and signal processing operations, Pandas for data handling, and Scikit-learn for machine-learning, feature selection, clustering, and subject-independent validation. XGBoost will be used for gradient-boosting-based classification, while Matplotlib may be used for visualization of physiological features and model results. The analysis can be conducted using Jupyter Notebook or a suitable Python development environment.

6. Expected Outcomes

The proposed study is expected to provide a systematic understanding of the complementary information contained in EEG and ECG/HRV signals for characterizing physiological differences associated with yoga experience. First, the study is expected to identify EEG features that differ between experienced yoga practitioners and non-practitioners across resting, internal concentration, external concentration, and mind-wandering conditions. Similarly, ECG-derived HRV features are expected to provide information regarding differences in cardiac autonomic activity between the two groups.

A major expected outcome is a direct comparison between EEG-only, ECG/HRV-only, and combined EEG–ECG representations. This comparison will help determine whether multimodal information provides an advantage over individual physiological modalities. The study will additionally investigate brain–heart coupling to determine

whether the relationship between neural and cardiac activity contains information beyond conventional modality-specific features and whether this relationship varies across cognitive states.

The machine-learning component is expected to identify which modelling approaches most effectively capture these physiological differences. Logistic Regression, SVM, Random Forest, and XGBoost will be compared using subject-independent validation, providing a more reliable estimate of model generalization. The analysis is also expected to reveal whether physiological differences associated with yoga experience are consistent across cognitive conditions or are primarily state-dependent.

Finally, K-Means clustering is expected to provide an unsupervised view of the data by identifying potentially distinct physiological profiles without using practitioner labels during cluster formation. Together, these outcomes may establish whether integrating neural activity, cardiac regulation, and their interaction provides a more informative representation of yoga-associated physiological patterns in terms of mental health and cognitive functioning.

7. Conclusion

This study proposes a multimodal framework for investigating physiological differences between experienced yoga practitioners and non-practitioners using synchronized EEG and ECG recordings. Existing research has demonstrated the individual relevance of neural and cardiac measures as well as the potential benefits of multimodal physiological analysis, but the interaction between these systems remains comparatively less explored within yoga-specific populations.

The proposed work addresses this opportunity by systematically comparing EEG-only, ECG/HRV-only, and combined EEG–ECG representations while additionally examining brain–heart coupling across different cognitive states. The use of multiple machine-learning models and subject-independent validation will provide a rigorous assessment of the ability of these representations to capture yoga-associated physiological differences. K-Means clustering will further provide an unsupervised perspective on the underlying physiological profiles.

Overall, the study aims to move beyond analyzing brain and cardiac activity independently and investigate whether their combination and interaction can provide a more comprehensive understanding of physiological differences associated with yoga experience in terms of mental health and cognitive functioning.

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